

LATROBE VALLEY, Australia

WMO: 948910

Lat: 38.2094S

Lon: 146.4747E

Elev: 56

StdP: 100.65

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.1	1.3	-0.7	3.6	2.0	0.4	3.9	3.5	12.8	10.0	11.5	10.7	0.7	230	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	33.7	19.5	30.9	19.2	28.4	18.6	21.3	28.7	20.3	27.4	19.5	26.1	4.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.1	23.0	18.2	13.2	22.2	17.3	12.5	21.5	62.0	28.7	58.8	27.8	55.7	26.1	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
11.7	10.3	9.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-1.9	39.3	0.8	2.2	-2.5	40.9	-3.0	42.2	-3.4	43.4	-4.0	45.0
			-2.1	23.5	0.9	1.9	-2.8	24.9	-3.3	26.0	-3.8	27.0	-4.4	28.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	19.8	19.6	17.6	14.4	11.5	9.2	8.7	9.4	11.2	13.2	15.6	17.5	(6)
		DBStd	4.81	3.77	3.41	3.36	2.77	2.16	2.07	1.96	2.04	2.32	2.89	3.27	3.52	
		HDD10.0	151	0	0	0	2	9	39	50	35	12	4	1	0	
		HDD18.3	1818	27	23	55	120	211	273	299	276	215	163	96	58	
		CDD10.0	1589	303	268	236	135	56	16	9	17	47	102	168	234	
		CDD18.3	216	72	58	33	3	0	0	0	0	0	2	13	34	
		CDH23.3	2319	730	517	307	49	1	0	0	0	7	78	207	422	
		CDH26.7	942	337	211	109	9	0	0	0	0	1	19	71	185	
(14)	Wind	WSAvg	3.8	4.3	4.0	3.8	3.2	3.4	3.1	3.6	3.9	4.2	4.1	4.3	4.4	(14)
(15)	Precipitation	PrecAvg	760	53	36	51	56	67	63	66	72	80	73	67	63	(15)
		PrecMax	983	151	135	204	119	177	128	134	152	160	174	131	196	
		PrecMin	466	0	2	5	7	10	15	14	24	27	26	24	6	
		PrecStd	130	29	30	37	31	39	25	32	32	31	34	31	39	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	35.7	33.7	27.7	22.0	17.7	17.7	19.8	24.0	29.1	32.1	36.4	(19)
			MCWB	20.5	19.6	18.7	16.3	14.4	12.4	11.4	13.2	15.1	17.2	19.1	20.0	
		2%	DB	33.8	32.0	29.7	24.4	19.3	15.6	15.5	17.3	20.7	25.3	28.6	31.7	
			MCWB	19.9	19.5	18.3	15.9	13.9	11.6	10.9	11.8	14.0	16.1	18.4	19.0	
		5%	DB	30.2	29.1	26.7	22.0	17.4	14.3	14.0	15.6	18.4	22.4	25.4	28.1	
			MCWB	19.5	19.3	17.8	15.4	13.0	11.1	10.3	10.9	13.1	15.2	17.2	18.1	
		10%	DB	27.1	26.3	24.0	19.7	15.9	13.2	12.8	14.0	16.3	19.6	22.4	24.7	
			MCWB	18.6	18.5	17.1	14.5	12.4	10.6	9.7	10.1	12.0	14.2	16.2	17.2	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	22.4	20.8	18.3	15.9	14.0	12.7	14.3	16.6	18.9	20.9	21.3	(27)
			MCDB	31.4	30.2	27.7	23.6	19.3	15.3	15.3	18.1	21.2	25.3	28.8	29.7	
		2%	WB	21.5	21.1	19.6	17.0	14.8	12.7	11.6	12.6	14.8	17.0	19.0	20.0	
			MCDB	29.2	27.9	26.0	21.5	17.5	14.6	14.2	16.0	19.3	23.3	26.5	27.9	
		5%	WB	20.4	20.1	18.6	16.1	13.8	11.7	10.7	11.4	13.5	15.7	17.9	19.0	
			MCDB	27.8	26.2	24.3	20.2	16.6	13.6	13.2	14.5	17.5	20.8	23.7	26.0	
		10%	WB	19.3	19.1	17.7	15.1	12.9	10.9	10.1	10.5	12.4	14.5	16.7	17.8	
			MCDB	26.0	24.8	22.9	18.9	15.4	12.8	12.3	13.3	15.7	19.2	21.6	23.8	
(35)	Mean Daily Temperature Range	MDBR		12.8	12.1	11.5	10.7	9.1	8.5	8.6	9.4	10.1	11.3	11.7	12.3	(35)
		5% DB		MCDBR	18.8	17.5	16.5	15.0	12.1	10.2	11.1	12.8	14.7	17.9	18.3	
				MCWBR	6.7	6.4	6.2	7.0	7.2	6.8	7.2	8.6	9.4	8.7	7.8	
		5% WB		MCDBR	15.4	14.1	13.5	12.1	10.8	9.1	9.3	11.4	13.1	15.2	15.2	
				MCWBR	6.3	5.9	5.8	6.1	6.9	6.5	6.4	7.4	8.1	8.6	7.9	
(40)	Clear-Sky Solar Irradiance	taub		0.375	0.368	0.345	0.334	0.318	0.302	0.317	0.313	0.347	0.346	0.351	0.363	(40)
		taud		2.421	2.461	2.517	2.527	2.550	2.589	2.524	2.536	2.433	2.461	2.473	2.448	
		Ebn at Noon		957	939	922	868	822	810	811	878	903	951	974	974	
		Edh at Noon		122	112	98	85	72	64	72	82	104	111	115	120	
(44)	All-Sky Solar Radiation	RadAvg		6.88	5.90	4.74	3.33	2.25	1.79	2.01	2.81	3.97	5.24	6.14	6.85	(44)
		RadStd		0.47	0.37	0.30	0.19	0.12	0.13	0.11	0.12	0.29	0.39	0.38	0.42	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+0.50	N/A	N/A	+0.79	N/A	-18	-130	+171	+61

Nomenclature: See separate page